

Kirak Kim

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Research Interests: Generative Models for Spatial Audio, Immersive Auditory Experience in VR

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST)

Daejeon, South Korea

Ph.D. Student, Graduate School of Culture Technology

Sep. 2025 –

Advisor: Sungyoung Kim

Korea Advanced Institute of Science and Technology (KAIST)

Daejeon, South Korea

M.S., Graduate School of Metaverse and Graduate School of Culture Technology

Sep. 2023 – Aug. 2025

Advisor: Juhan Nam

Sogang University

Seoul, South Korea

Completed Master's Coursework, Department of Art & Technology

Sep. 2021 – Aug. 2023

Advisor: Moonryul Jung

Yonsei University

Seoul, South Korea

B.S., School of Integrated Technology

Mar. 2018 – Feb. 2021

Advisor: Jongseok Lee

EXPERIENCE

Research Assistant

Sep. 2025 –

Applied and Innovative Research for Immersive Sound Lab, KAIST

- Leading the spatial audio & AI team, conducting research on AI-driven room acoustics modeling
- Designed an LLM/VLM-based pipeline to caption and curate spatial audio datasets for training a text-conditioned room impulse response (RIR) generation model
- Exploring multimodal LLM-based approaches for spatial audio generation

Visiting Researcher

Jan. 2026 – June. 2026

Centre for Analytics & AI Engineering, University of Toronto

- Conducting collaborative research with LG Toronto AI Lab on improving Vision-Language-Action (VLA) models
- Investigating Vision-Language Model (VLM) latent representations for long-horizon action prediction using probing and auxiliary loss objectives

Research Assistant

Sep. 2023 – Aug. 2025

Music and Audio Computing Lab, KAIST

- Developed a diffusion-based model with spatio-temporal graph neural networks to generate pianist hand motion from music, in collaboration with YAMAHA
- Collected multimodal piano performance datasets and contributed to audiovisual piano transcription research
- Participated in AI-driven piano performances and exhibitions at venues including Daejeon Arts Center

Research Assistant

Sep. 2021 – Aug. 2023

Korea Electronics Association

- Integrated AI models within virtual environments
- Built a real-time multimodal dialogue system generating emotion-based utterances and facial expressions for virtual humans
- Developed a real-time background music selection system for metaverse conversations using speech emotion regression

SELECTED PUBLICATIONS

Conference Papers

- **K. Kim**, and S. Kim. (2026). *Adapting a Text-to-Audio Model for Room Impulse Response Generation*. In Proceedings of the 19th International Workshop on Acoustic Signal Enhancement (IWAENC 2026).[paper]
- **K. Kim**, H. Kim, Y. Choi, J. Nam, and J. Lee. (2025). *VR Music Game for Stress Reduction*. In Proceedings of the IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR), pp. 678–685.[paper]
- Y. Kim, J. Park, J. Bae, **K. Kim**, T. Kwon, A. Lerch, and J. Nam. (2025). *PianoVAM: A Multimodal Piano Performance Dataset*. In Proceedings of the 26th International Society for Music Information Retrieval Conference (ISMIR).[paper]

Poster / Demo / Workshop Papers

- **K. Kim**, H. Kim, Y. Son, S. Kim, K. Lee. (2026). *Does Persona Make LLMs K-pop Fans? A Pilot Study of LLM-Based Online Concert Audience Agents*. ICML 2026 Workshop on Culture x AI: Evaluating AI as a Cultural Technology.[paper]
- **K. Kim**, H. Kim, S. Bae, J. Lee, S. Kim. (2025). *The Soaring Serenity: an Immersive VR Artwork for Perspective Shift*. 30th International Symposium on Electronic/Emerging Art (ISEA) Poster.[paper]
- J. Bae, H. Cho, **K. Kim**, D. Park, T. Kwon, Y. Choi, H. Hur, S. Kai, Y. Wada, S. Obata, A. Maezawa, J. Park, J. Park, J. Nam. (2025). *Designing a Multimodal Viewer for Piano Performance Analysis - a Pedagogy-First Approach*. Late Breaking Demo of the 26th International Society for Music Information Retrieval Conference (ISMIR).[paper]

HONORS & AWARDS

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| 2024 | 3rd Prize, AI x Art Hackathon, Seoul National University Institute for Culture and Art |
| 2023 | 2nd Prize, Korea Electronics Association AR/VR Specialist Training Program Research Symposium |
| 2022 | 1st Prize, Korea Electronics Association AR/VR Specialist Training Program Research Symposium |

ACADEMIC SERVICES

Conference Organizing

Volunteer Co-Chair, the 26th International Society for Music Information Retrieval Conference (ISMIR 2025)

Reviewing

IEEE IS2, IWAENC 2026, CHI 2026 Posters

TECHNICAL SKILLS

Programming: Python, C#

ML Frameworks: PyTorch, HuggingFace Transformers

Tools: Unity 3D, Wwise, Git

Research Skills: Multimodal Generative Modeling, HCI Evaluation